Bit Manipulation Study Guide

Revision on Bit Manipulation

What is Bit Manipulation?

Bit manipulation involves performing operations directly on binary representations of numbers. In Java, integers are stored as 32-bit signed values, and we can manipulate individual bits using bitwise operators.

Binary Number System Review

- **Binary**: Base-2 number system using only 0 and 1
- **Decimal to Binary**: Divide by 2 repeatedly and read remainders bottom-up
- **Binary to Decimal**: Multiply each bit by its corresponding power of 2

Example:

```java
// Decimal 13 in binary is 1101
// 1×2³ + 1×2² + 0×2¹ + 1×2⁰ = 8 + 4 + 0 + 1 = 13
```

Bitwise Operators in Java

1. AND Operator (&)

Returns 1 only when both bits are 1.

```java
int a = 5;  // Binary: 0101
int b = 3;  // Binary: 0011
int result = a & b;  // Binary: 0001 = 1
System.out.println(result); // Output: 1
```

2. OR Operator (|)

Returns 1 when at least one bit is 1.

```java
int a = 5;  // Binary: 0101
int b = 3;  // Binary: 0011
int result = a | b;  // Binary: 0111 = 7
System.out.println(result); // Output: 7
```

3. XOR Operator (^)

Returns 1 when bits are different.

```java
int a = 5;  // Binary: 0101
int b = 3;  // Binary: 0011
int result = a ^ b;  // Binary: 0110 = 6
System.out.println(result); // Output: 6
```

4. NOT Operator (~)

Inverts all bits (1s complement).

```java
int a = 5;  // Binary: 0101
int result = ~a;  // Inverts all 32 bits
System.out.println(result); // Output: -6
```

5. Left Shift (<<)

Shifts bits to the left, filling with zeros.

```java
int a = 5;  // Binary: 0101
int result = a << 2;  // Binary: 10100 = 20
System.out.println(result); // Output: 20
```

6. Right Shift (>>)

Shifts bits to the right, preserving sign bit.

```java
int a = 20;  // Binary: 10100
int result = a >> 2;  // Binary: 0101 = 5
System.out.println(result); // Output: 5
```

7. Unsigned Right Shift (>>>)

Shifts bits to the right, filling with zeros.

```java
int a = -8;
int result = a >>> 2;
System.out.println(result); // Positive result due to zero-fill
```

Common Bit Manipulation Techniques

Check if a Number is Even or Odd

```java
public static boolean isEven(int n) {
    return (n & 1) == 0;
}

public static boolean isOdd(int n) {
    return (n & 1) == 1;
}
```

Check if nth Bit is Set

```java
public static boolean isBitSet(int num, int position) {
    return (num & (1 << position)) != 0;
}
```

Set nth Bit

```java
public static int setBit(int num, int position) {
    return num | (1 << position);
}
```

Clear nth Bit

```java
public static int clearBit(int num, int position) {
    return num & ~(1 << position);
}
```

Toggle nth Bit

```java
public static int toggleBit(int num, int position) {
    return num ^ (1 << position);
}
```

Count Number of Set Bits

```java
public static int countSetBits(int n) {
    int count = 0;
    while (n != 0) {
        count++;
        n = n & (n - 1);  // Brian Kernighan's algorithm
    }
    return count;
}
```

Check if Number is Power of 2

```java
public static boolean isPowerOfTwo(int n) {
    return n > 0 && (n & (n - 1)) == 0;
}
```

Find the Only Non-Duplicate Number

```java
public static int findSingle(int[] arr) {
    int result = 0;
    for (int num : arr) {
        result ^= num;  // XOR cancels out duplicates
    }
    return result;
}
```

Bit Masking

Bit masking is used to extract specific bits from a number or to modify certain bits.

```java
// Extract last 4 bits
int num = 157;  // Binary: 10011101
int mask = 15;  // Binary: 00001111
int lastFourBits = num & mask;  // Result: 13 (1101)

// Set multiple bits using mask
int setBits = num | mask;

// Clear multiple bits using mask
int clearBits = num & ~mask;
```

Two's Complement

Java uses two's complement representation for negative numbers:

- Positive numbers: Normal binary representation
- Negative numbers: Invert all bits and add 1

```java
int positive = 5;   // Binary: 00000101
int negative = -5;  // Binary: 11111011 (two's complement)
```

Quiz Questions

1. What is the result of 5 & 3? a) 5
b) 3
c) 1
d) 7

Answer: c) 1

2. Which operator is used to check if a number is even? a) |
b) &
c) ^
d) ~

Answer: b) &

3. What does the expression (n & (n-1)) == 0 check for? a) If n is even
b) If n is odd
c) If n is a power of 2
d) If n is negative

Answer: c) If n is a power of 2

4. What is the result of 8 >> 2? a) 32
b) 4
c) 2
d) 16

Answer: c) 2

5. Which operation can be used to find a single non-duplicate number in an array where all other numbers appear twice? a) AND
b) OR
c) XOR
d) NOT

Answer: c) XOR

Practice Coding Questions

1. Write a Java method that swaps two integers without using a temporary variable, using only bitwise operations.

2. Create a function that finds the position of the rightmost set bit in a given integer. For example, if the input is 12 (binary: 1100), the output should be 2.

3. Implement a method that reverses the bits of a 32-bit integer. For example, if the input is 43261596 (binary: 00000010100101000001111010011100), the output should be 964176192.

4. Write a program that finds two non-repeating numbers in an array where every other number appears exactly twice.

5. Create a function that checks if a given number can be expressed as a sum of two powers of 2. Return true if possible, false otherwise.